

NIBB ゲノムインフォマティクス・トレーニングコース2024
RNA-seq入門
2024.7.31-2024.8.1

NGS可視化ツールIGV

基礎生物学研究所
超階層生物学センター
トランスオミクス解析室
山口勝司



データ可視化ツール・IGVの紹介・実習

IGV Integrative Genomics Viewer



IGV

IGV desktop application



IGV Web App

IGV in a web browser



Juicebox Web

Hi-C contact map viewer

<https://igv.org>

For Developers

igv.js

Use igv.js to embed an interactive genome visualization component in your web app.

igv-notebook

A Python package that wraps igv.js for embedding in an IPython notebook. Supports both Jupyter and Google Colab.

igv-reports

Generate self-contained HTML reports that consist of a table of genomic sites and associated IGV views for each site.

All IGV software is open source - MIT License.

GitHub repos: [IGV Desktop](#) | [IGV Web App](#) | [igv.js](#) | [igv-notebook](#) | [igv-reports](#) | [Juicebox Web](#) | [juicebox.js](#)

Citing IGV

To cite your use of IGV in your publication, please reference one or more of:

James T. Robinson, Helga Thorvaldsdóttir, Wendy Winckler, Mitchell Guttman, Eric S. Lander, Gad Getz, Jill P. Mesirov. Integrative Genomics Viewer. *Nature Biotechnology* 29, 24–26 (2011). A public access version is also available: [PMC3346182](#).

Helga Thorvaldsdóttir, James T. Robinson, Jill P. Mesirov. Integrative Genomics Viewer (IGV): high-performance genomics data visualization and exploration. *Briefings in Bioinformatics* 14, 178-192 (2013).

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Contact Us

The IGV Team is based at UC San Diego and the Broad Institute of MIT and Harvard. The best way to reach us for support questions, bug reports, feature requests, and suggestions is by posting to the [igv-help](#) forum or by creating new issues in our GitHub repositories. To reach us privately on other topics not related to general support, feel free to email us directly.

Funding

IGV development has been supported by funding from the National Cancer Institute (NCI) of the National Institutes of Health, the Informatics Technology for Cancer Research (ITCR) of the NCI, and the Starr Cancer Consortium.

IGV desktop



IGV desktop application

IGV Desktop Application

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About IGV



<https://igv.org/doc/desktop>

Overview

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なぜIGVを取り上げるか

データ可視化ツール 2つのタイプ

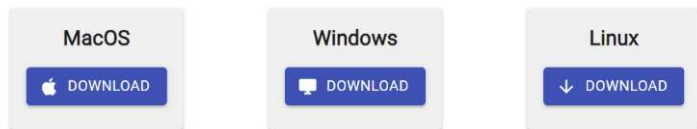
- ・自分のパソコン(ローカル環境)にインストールして使うタイプ
- ・サーバーに構築して、ネットワーク上で使うタイプ

Download JBrowse 2

JBrowse 2 is a new browser for today's increasingly complex data. We make several different products for use in different scenarios, choose the one below that best fits your needs.

JBrowse Desktop

A standalone app that runs on your own computer that can view data from many sources. Download JBrowse Desktop on the platform of your choice:



Check out our quick start guide [instructions after downloading](#). Can't find what you're looking for? Checkout [other platforms](#).

JBrowse Web

Run a full-featured genome browser with many types of built-in views on your own website.



後者はコミュニティでの利用目的、ウェブ公開を目的とするには良い。より大容量なデータに対応できる。

ネットワーク・情報セキュリティに対する高度な知識が要求される。

管理者的な人がいるなら、これも良いが・・・

JBrowseも現在はIGV的な利用も可

JBrowse desktop

もっとお手軽なものとしてIGVを紹介

可視化ツールに求められるものは何か

膨大なデータを、如何に直感的に理解できるか
sortや絞り込みで客観的に理解する表データと対比双璧になる

多様なデジタル情報

- ・ 配列、GC ratio、遺伝子情報
- ・ 遺伝子発現情報
- ・ 遺伝子多型の位置情報・頻度情報
- ・ 様々なデータの精度情報
- ・ RNA-seq, ChIP-seq, RAD-seq, BS-seq . . .

レファレンス配列 / gene model / gene annotationと
NGSデータを並べて比較
複数のデータセットを並べて比較

様々なスケールで比較・統合的に解釈できるようにしたい

ゲノムviewerに自分のデータを乗せ、
統合的直感的に比較・解釈できること

可視化ツールをどう選ぶか

選択の基準

genome data viewing に求められるもの

取捨選択の基準

1. 無料 / 有料 / 基本無料
2. 個人的レベルの使用 / コミュニティーレベルの使用
3. 見るだけ/自分から色々工夫
4. アクセシビリティ・ユーザビリティ
 - 導入に必要なコンピュータスペック
 - マニュアルは分かりやすいか
 - 情報の多さ
 - 利用の簡便さ
 - 使っている人が近くにいますか

Integrative Genomics Viewer(IGV)



お手軽ツール

- 無料
- 多くの人が使っている、情報も多い
- javaのプログラムなので、オールプラットフォーム対応
- マニュアルは親切、サンプルデータもある
- WEBサーバーではなく、PCレベルで利用出来る
- データ閲覧環境の共有も可能

誰もが簡便に使えるものが良い。

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Integrative genomics viewer

James T Robinson, Helga Thorvaldsdóttir, Wendy Winckler, Mitchell Guttman, Eric S Lander, Gad Getz & Jill P Mesirov

[Affiliations](#) | [Corresponding authors](#)

Nature Biotechnology 29, 24–26 (2011) | doi:10.1038/nbt.1754

Published online 10 January 2011

To the Editor:

Rapid improvements in sequencing and array-based platforms are resulting in a flood of diverse genome-wide data, including data from exome and whole-genome sequencing, epigenetic surveys, expression profiling of coding and noncoding RNAs, single nucleotide polymorphism (SNP) and copy number profiling, and functional assays. Analysis of these large, diverse data sets holds the promise of a more comprehensive understanding of the genome and its relation to human disease. Experienced and knowledgeable human review is an essential component of this process, complementing computational approaches. This calls for efficient and intuitive visualization tools able to scale to very large data sets and to flexibly integrate multiple data types, including clinical data. However, the sheer volume and scope of data pose a significant challenge to the development of such tools.

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Citations to this article

[Crossref \(10\)](#) [Scopus \(12\)](#) [Web of Science \(0\)](#)

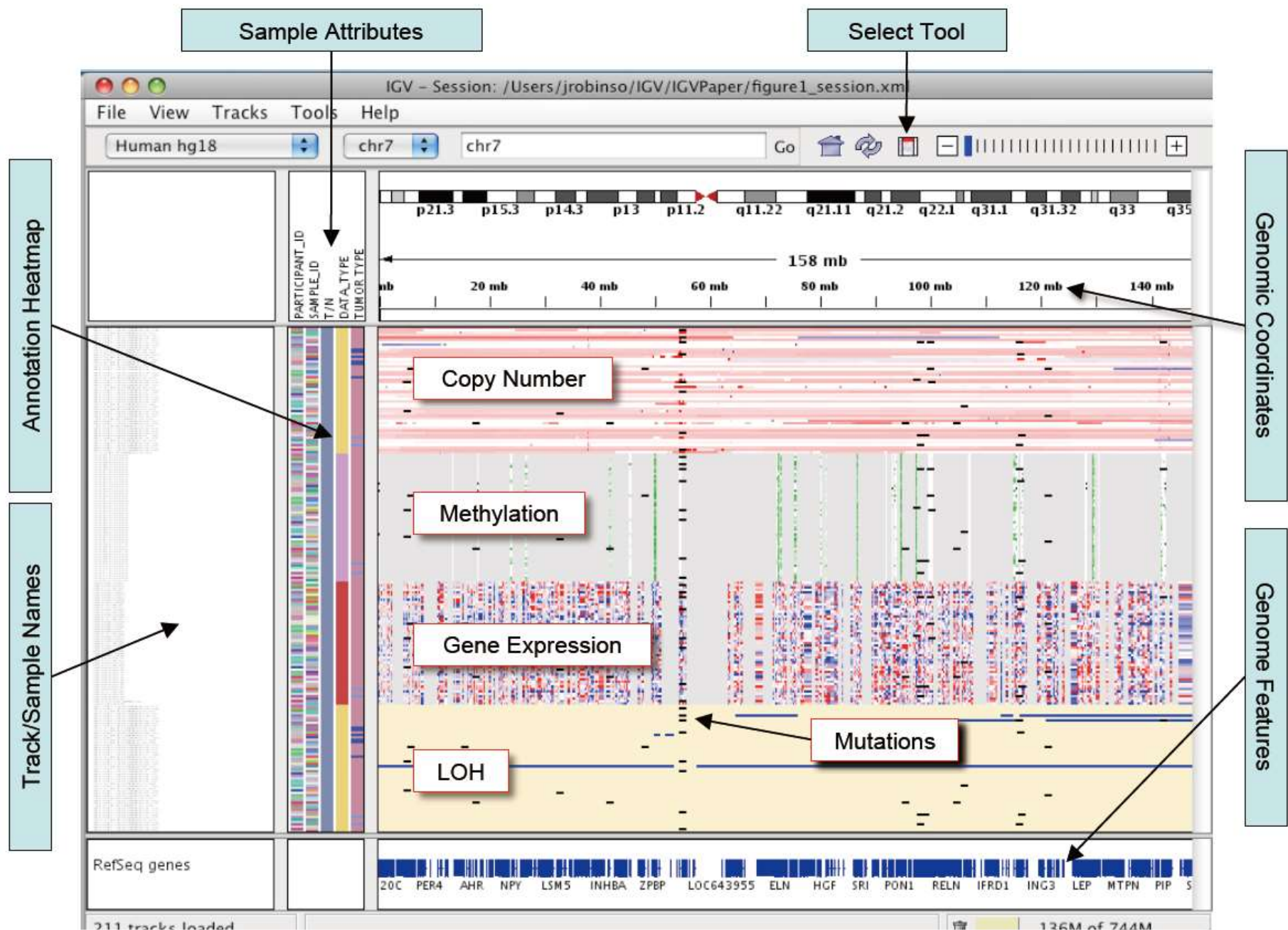
Science jobs from [naturejobs](#)

Faculty Position

Harvard Medical School

Ramalingaswami Re-Entry Fellowship

Ministry of Science & Technology, Government of India



Nature Biotech. 29:24–26 (2011) Supplement figureからの抜粋

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 - broadPeak
 - CBS
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 - GWAS
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 - LOH
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 - Merged BAM File
 - narrowPeak
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Download IGV

Did you know that there is also an **IGV web application** that runs only in a web browser, does not use Java, and requires no downloads? See <https://igv.org/app>. Click on the Help link in the app for more information about using IGV-Web.

Latest release of IGV Desktop: 2.17.4

As of version 2.17.0, IGV requires Java 17 or greater.

If you download one of the IGV versions that does not include Java, make sure you have Java 17 installed and in your path.

M1/M2 Mac Users:

Apple's Rosetta software is required to run the IGV MacOS App that includes Java. If you run IGV with your own Java installation, Rosetta may not be required if your version of Java runs natively on M1/M2.

Linux users:

The 'IGV for Linux' download includes AdoptOpenJDK (now Eclipse Temurin) version 17 for x64 Linux. See their list of supported platforms [here](#). If your platform is not on the "x64 Linux" list, or the packaged Java does not work on your version of Linux, download the 'Command line IGV for all platforms' and use it with your own Java installation.

What's New: See the [Release Notes](#) for what's new in each IGV release.



IGV MacOS App
Java included



IGV MacOS App
Separate Java 17 required



IGV for Windows
Java included



IGV for Windows
Separate Java 17 required



IGV for Linux
Java included



Command line IGV and igvtools for all platforms
Separate Java 17 required

利用するパソコンの
プラットフォームに応じて選択

Other releases of IGV Desktop

Development snapshot build. Latest development snapshot; built at least nightly

Archived releases. Old releases going back to IGV 2.0

Other versions of IGV

This Downloads page is for the IGV desktop version. See also:

- If you are looking for the IGV-Web application, see <https://igv.org/app>
- If you are a developer looking for information about the embeddable igv.js component, see <https://github.com/igvteam/igv.js>
- If you are developer wanting to use igv.js in a notebook (Jupyter or Colab), see <https://github.com/igvteam/igv-notebook>
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IGV

File Genomes View Tracks Regions Tools Help

Select Hosted Genome...

Load Genome from File...

Load Genome from URL...

Load Track Hub...

Select Hub Tracks...

Remove Genomes...

Select genomes available on the server to appear in menu.

Hosted Genomes

Selected genomes will be downloaded and added to the genome dropdown list.

Filter:

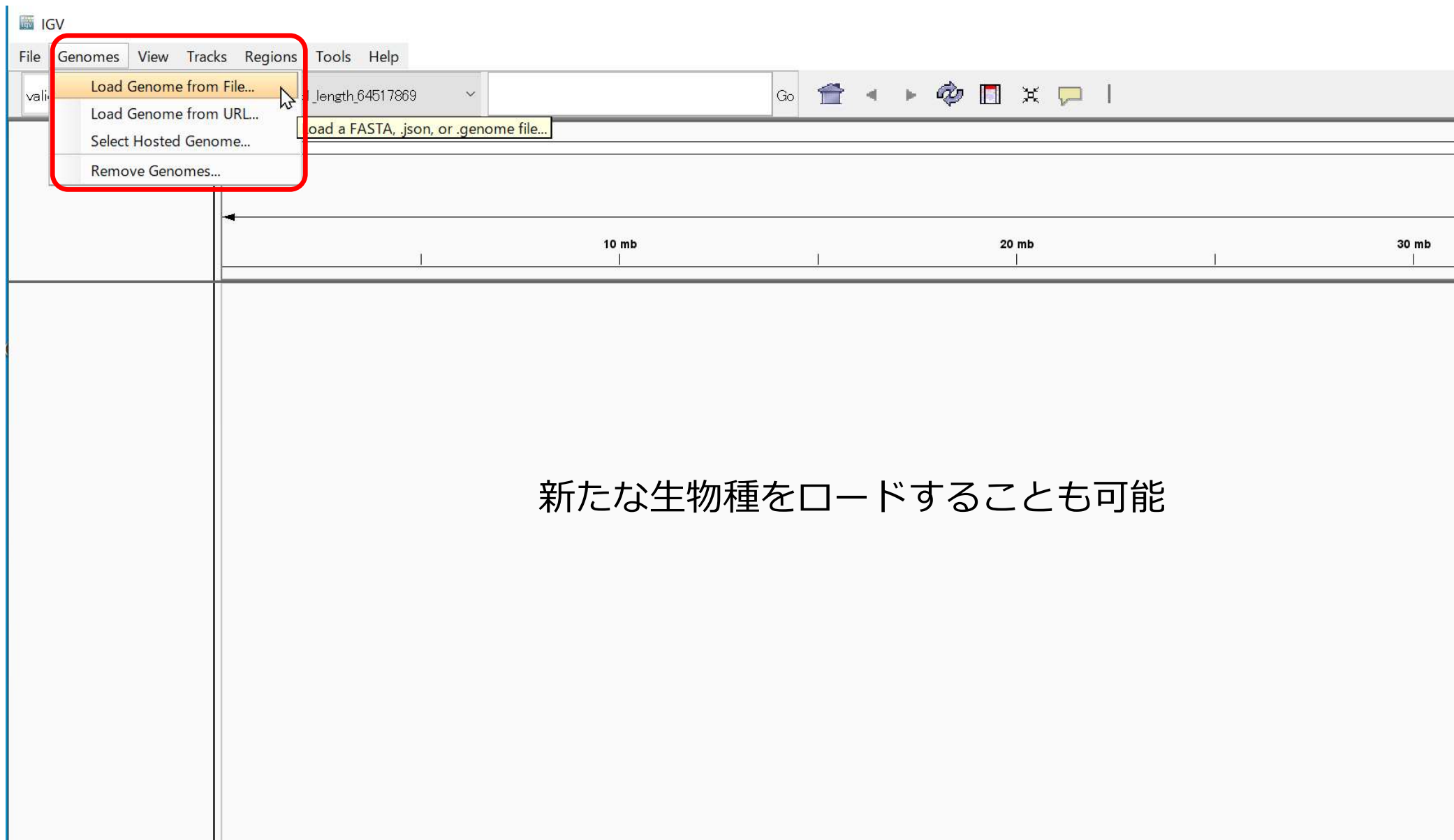
- A. gambia (Pest AgamP3)
- A. japonica (Ajap1.0)
- A. nidulans (4.1)
- A. planci (Aplan1.0)
- A. rubens (Arub1.3)
- A. thaliana (TAIR 10)
- Autographa californica MNPV
- Bacillus Subtilis str. 168
- Banana (M. Balbisiana PKWv1)
- Banana (Musa acuminata)
- Bonobo (MPI-EVA panpan1.1/panPan2)
- C. alibicans (SC5314 A21)
- C. elegans (ce10)
- C. elegans (ce11)
- C. elegans (WS235)
- C. elegans (WS241)
- C. elegans (WS245)
- C. reinhardtii (CC-503 v5.5)
- C. reinhardtii (CC-503 v5.6 Cp+Mt v4.4)
- C. reinhardtii (CC-503 v5.6)
- Cat (felCat5)
- Chicken (galGal4)
- Chimpanzee (chr1)

☐ Download Sequence

OK Cancel

いくつかの生物種は登録されている

231M of 616M



新たな生物種をロードすることも可能

ゲノムViewerなので次世代DNAシーケンサーのデータに限定されない。
マイクロアレイの結果や、ゲノムアノテーションの情報も随時表示できる。

対応するファイル形式に応じて、表示方法が決まる。

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File Formats

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- [TDF](#)
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- [Type Line](#)
- [VCF](#)
- [WIG](#)

File Formats

IGV supports a number of different file formats for experimental data and genome annotations. For a complete list of supported formats see <http://www.broadinstitute.org/igv/FileFormats>. The following table shows the recommended file formats for a number of common data types.

Source Data	Recommended File Formats
ChIP-Seq, RNA-Seq	WIG, TDF
Copy number	CN, SNP, TDF, canary_calls (Birdsuite)
Gene expression data	GCT, RES, TDF
Genome annotations	GFF, BED, GTF, PSL, UCSC table format
GISTIC data	GISTIC
LOH data	LOH, TDF
Mutation data	MUT, MAF
Variant calls	VCF
RNAi data	GCT
Segmented data	SEG, CBS
Sequence alignment data	BAM, SAM, PSL
Any numeric data	IGV, WIG, TDF
Sample metatadata	Tab-delimited sample info file

その他の便利機能

セッションの保存

表示しているデータの読み込み状況を、それごと保存。

セッションをロードすることで、意図した画面を表示できる。

データセットが揃っていること、フォルダー構造が同一である必要がある。

Ref配列のコピー

Define a region of interestのアイコンを押す。

カーソルが十字となる。

配列の取り出したい領域の開始と終了ポジションをクリック。

指定された領域が赤色になり、右クリックしてCopy Sequence

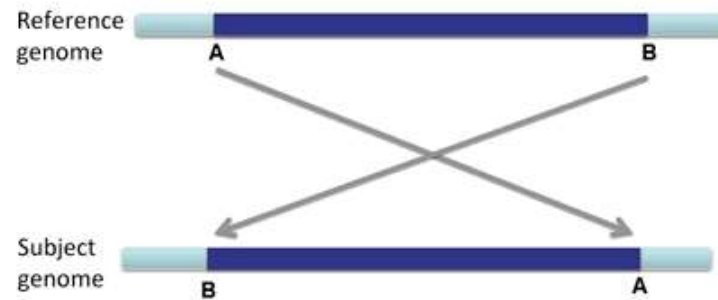
バッチ処理

重要領域の画面スナップショットを自動で取ったりできる。

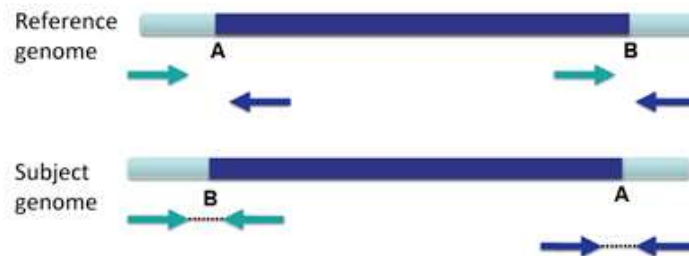
```
new
load myfile.bam
snapshotDirectory mySnapshotDirectory
genome hg18
goto chr1:65,289,335-65,309,335
sort position
collapse
snapshot
goto chr1:113,144,120-113,164,120
sort base
collapse
snapshot
```

Inversions

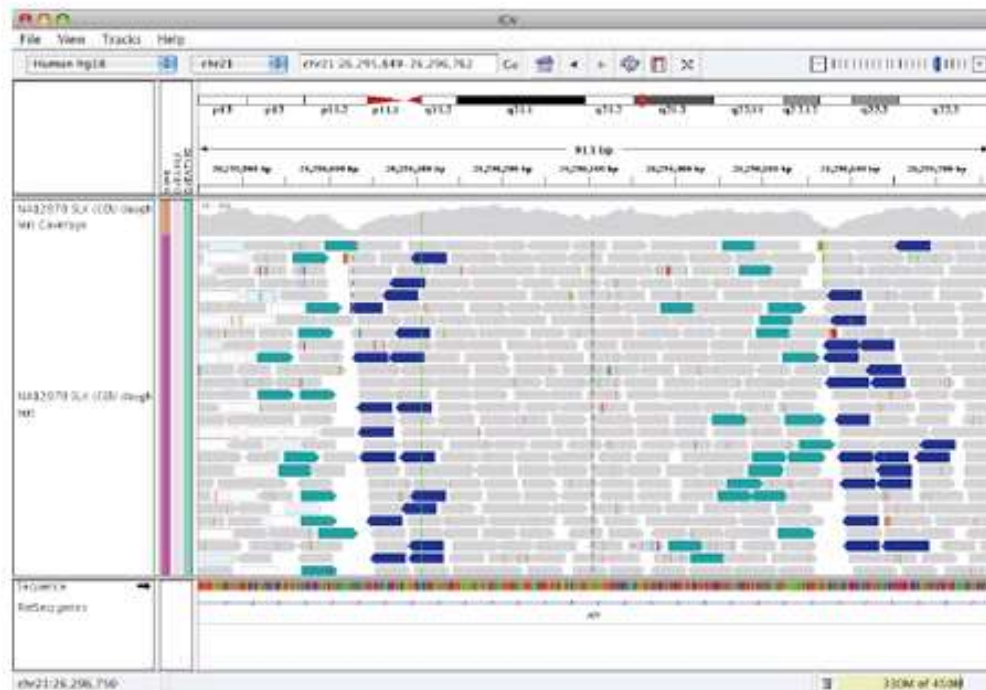
An inversion is a large section of DNA that is reversed in the subject genome compared to the reference genome.



When an inversion shows up in paired-end reads, the reads are distinctively variant from the reference genome.



This appears in IGV as shown below.



Interpreting Color by Insert Size

The inferred insert size can be used to detect structural variants, such as:

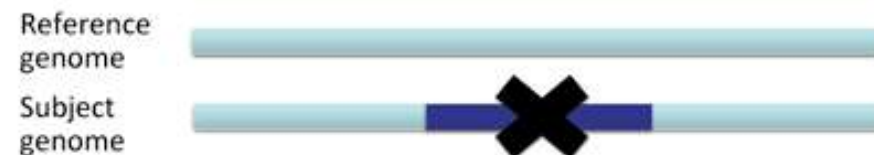
- deletions
- insertions
- inter-chromosomal rearrangements

IGV uses color coding to flag anomalous insert sizes. When you select Color alignments>by insert size in the popup menu, the default coloring scheme is:

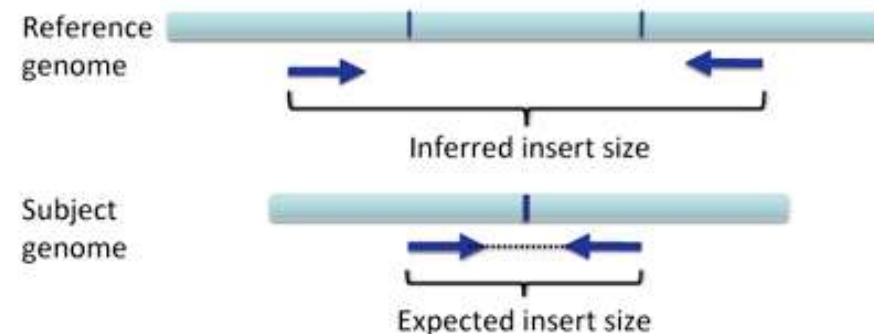
-  for an insert that is larger than expected
-  for an insert that is smaller than expected
-  for paired end reads that are coded by the chromosome on which their mates can be found

Deletions

A deletion is a large section of DNA that is absent in the subject genome compared to the reference genome.



The "expected" insert size is the insert size obtained in sequencing the subject genome. The "inferred" insert size is the insert size that would result in the reference genome, assuming the same pair of reads.

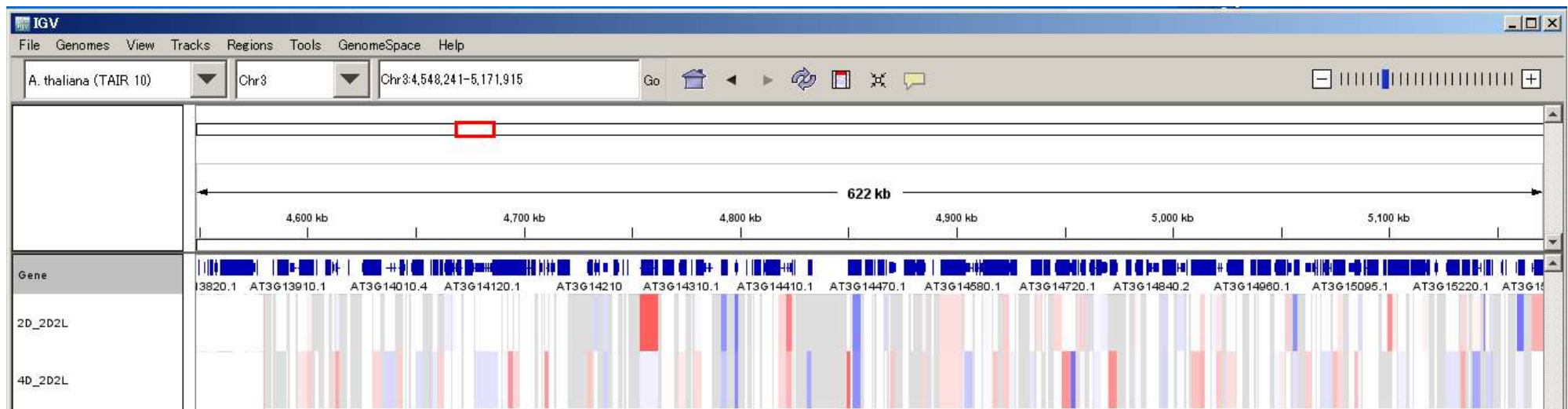


RNA-Seqのデータ表示させる



GCTファイルでgene ローカスの発現情報を図示

#				
#	Name	Description	2D_2D2L	4D_2D2L
	ANAC001	@Chr1:3630-5899	-2.60184	-2.60956
	DCL1	@Chr1:23145-33153	-0.742675	-1.5642
	MIR838A	@Chr1:23145-33153	0	0
	AT1G01073	@Chr1:44676-44787	0	0
	IQD18	@Chr1:52238-54692	-1.93871	-1.13128
	AT1G01115	@Chr1:56623-56740	0	0
	GIF2	@Chr1:72338-74737	-0.251287	-0.616679
	AT1G01180	@Chr1:75582-76758	0.45929	-0.809567
	AT1G01210	@Chr1:88897-89745	1.6964	0.857196
	FKGP	@Chr1:91375-95651	-0.174589	0.725947
	AT1G01240	@Chr1:99893-101834	-0.226384	-0.936641
	AT1G01260	@Chr1:108945-111609	-0.161848	0.315699
	CYP703A2	@Chr1:112262-113947	0	0
	CNX3	@Chr1:114285-116108	0.111249	-0.551359
	AT1G01300	@Chr1:116942-118764	-0.68348	0.108578



Gene listを定義して
サンプルごと
条件ごと
の発現・発現変動を
カラーマップできる

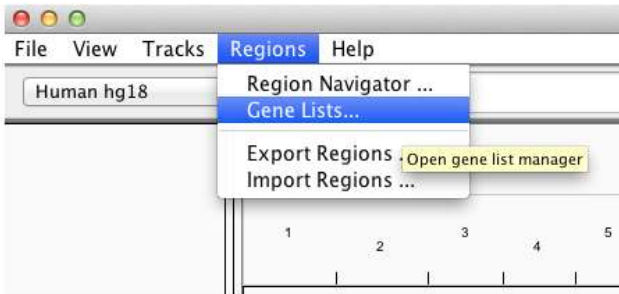
Home » IGV User Guide » Gene List View

Gene List View

The Gene Lists functionalities in IGV allow you to view lists of genes or loci side-by-side irrespective of their genomic location.

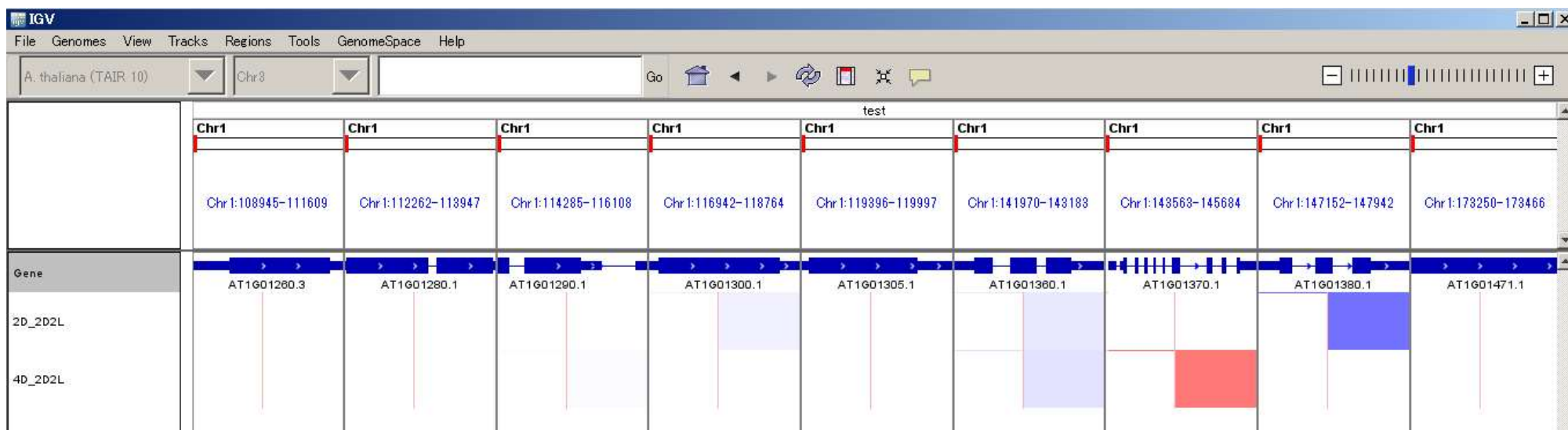
Loading/Defining Gene Lists

To load or define a new gene/locus list, select *Regions > Gene Lists...*



The screenshot shows the IGV application window. The menu bar at the top includes 'File', 'View', 'Tracks', 'Regions', and 'Help'. The 'Regions' menu is open, displaying options: 'Region Navigator ...', 'Gene Lists...', 'Export Regions', and 'Import Regions ...'. The 'Gene Lists...' option is highlighted with a blue background. A yellow tooltip with the text 'Open gene list manager' is visible next to the 'Export Regions' option. Below the menu, the 'Tracks' panel shows 'Human hg18'. The main display area shows a genomic track with five numbered regions (1, 2, 3, 4, 5) marked by vertical lines.

This opens a window for selecting an existing list or creating a new list.



IGV実習

IGV Desktop Application



- About IGV
- Quick Start
- User Guide
- File Formats
- Tutorial Videos
- Download IGV ▾
 - Latest release of IGV Desktop: 2.17.4
 - Other releases of IGV Desktop
 - Other versions of IGV
- Release Notes
- Getting Help

Download IGV

Did you know that there is also an **IGV web application** that runs only in a web browser, does not use Java, and requires no downloads? See <https://igv.org/app>. Click on the Help link in the app for more information about using IGV-Web.

Latest release of IGV Desktop: 2.17.4

As of version 2.17.0, IGV requires Java 17 or greater.

If you download one of the IGV versions that does not include Java, make sure you have Java 17 installed and in your path.

M1/M2 Mac Users:

Apple's Rosetta software is required to run the IGV MacOS App that includes Java. If you run IGV with your own Java installation, Rosetta may not be required if your version of Java runs natively on M1/M2.

Linux users:

The 'IGV for Linux' download includes AdoptOpenJDK (now Eclipse Temurin) version 17 for x64 Linux. See their list of supported platforms [here](#). If your platform is not on the "x64 Linux" list, or the packaged Java does not work on your version of Linux, download the 'Command line IGV for all platforms' and use it with your own Java installation.

What's New: See the [Release Notes](#) for what's new in each IGV release.



IGV MacOS App
Java included



IGV MacOS App
Separate Java 17 required



IGV for Windows
Java included



IGV for Windows
Separate Java 17 required



IGV for Linux
Java included



Command line IGV and igvtools for all platforms
Separate Java 17 required

Other releases of IGV Desktop

Development snapshot build. Latest development snapshot; built at least nightly

Archived releases. Old releases going back to IGV 2.0

Other versions of IGV

This Downloads page is for the IGV desktop version. See also:

- If you are looking for the IGV-Web application, see <https://igv.org/app>
- If you are a developer looking for information about the embeddable igv.js component, see <https://github.com/igvteam/igv.js>
- If you are developer wanting to use igv.js in a notebook (Jupyter or Colab), see <https://github.com/igvteam/igv-notebook>
- To generate self-contained HTML reports that consist of a table of genomic sites and associated IGV views, see <https://github.com/igvteam/igv-reports>

IGVの使用法を学ぶと共に
先のファイルフォーマットも
確認しよう

以下のファイルを確認

buc.genome.fasta

buc.gtf

buc_cg.wig

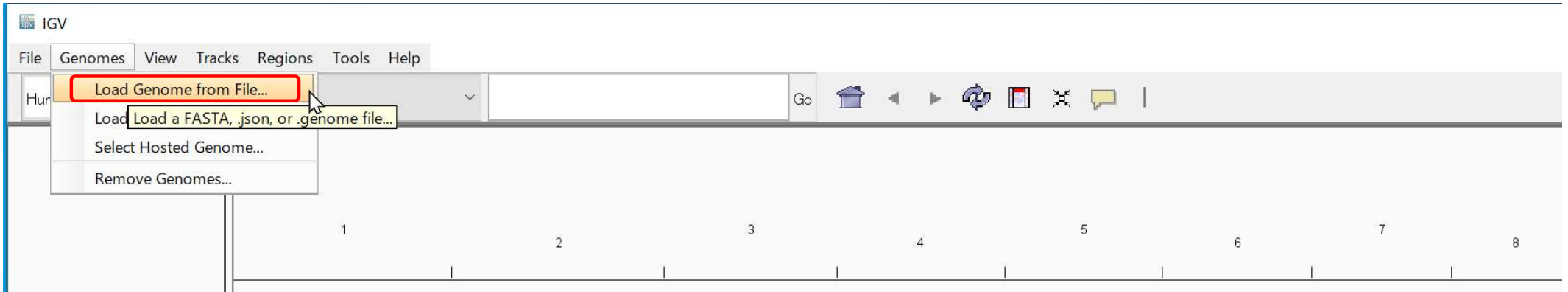
illumina_ex_B2_Read_bowtie2.mate.sort.bam

illumina_ex_B2_Read_bowtie2.mate.sort.bam.bai

illumina_ex_B4_Read_bowtie2.mate.sort.bam

illumina_ex_B4_Read_bowtie2.mate.sort.bam.bai

共生バクテリア 2系統を比較する



登録されていない生物種・配列でも、自分でimportすればOK

Buchneraゲノムを使う

Genome -> Load Genome from File
buc.genome.fasta
を読み込む

IGV

File Genomes View Tracks Regions Tools Help

Load from File... Load from URI Load tracks or sample information Load from Server... Reload Tracks New Session... Open Session... Save Session... Reload Session Save PNG Image ... Save SVG Image ... Exit

ゲノム構造を記述したgtfファイルを読み込む

File-> Load from File
buc.gtf ファイルを読み込む

3 4

buc.gtf

gn:buc	KEGG:GENES	CDS	197	2083	+	geneid "BU001"; transcript_id "BU001-RA"; gene_name "gidA";
gn:buc	KEGG:GENES	CDS	2278	3102	+	geneid "BU002"; transcript_id "BU002-RA"; gene_name "atpB";
gn:buc	KEGG:GENES	CDS	3139	3378	+	geneid "BU003"; transcript_id "BU003-RA"; gene_name "atpE";
gn:buc	KEGG:GENES	CDS	3497	3982	+	geneid "BU004"; transcript_id "BU004-RA"; gene_name "atpF";
gn:buc	KEGG:GENES	CDS	3982	4515	+	geneid "BU005"; transcript_id "BU005-RA"; gene_name "atpH";
gn:buc	KEGG:GENES	CDS	4530	6068	+	geneid "BU006"; transcript_id "BU006-RA"; gene_name "atpA";
gn:buc	KEGG:GENES	CDS	6101	6973	+	geneid "BU007"; transcript_id "BU007-RA"; gene_name "atpG";
gn:buc	KEGG:GENES	CDS	6997	8394	+	geneid "BU008"; transcript_id "BU008-RA"; gene_name "atpD";
gn:buc	KEGG:GENES	CDS	8421	8837	+	geneid "BU009"; transcript_id "BU009-RA"; gene_name "atpC";
gn:buc	KEGG:GENES	CDS	8911	11322	-	geneid "BU010"; transcript_id "BU010-RA"; gene_name "gyrB";
gn:buc	KEGG:GENES	CDS	11449	12549	-	geneid "BU011"; transcript_id "BU011-RA"; gene_name
"dnaN";						
gn:buc	KEGG:GENES	CDS	12554	13918	-	geneid "BU012"; transcript_id "BU012-RA"; gene_name "dnaA";
gn:buc	KEGG:GENES	CDS	14369	14512	+	geneid "BU013"; transcript_id "BU013-RA"; gene_name
"rpmH";						
gn:buc	KEGG:GENES	CDS	14525	14872	+	geneid "BU014"; transcript_id "BU014-RA"; gene_name "rnpA";
gn:buc	KEGG:GENES	CDS	15011	16609	+	geneid "BU015"; transcript_id "BU015-RA"; gene_name "yidC";

IGV

File Genomes View Tracks Regions Tools Help

buc.genome.fasta All

Go

gn:buc

1 track scaffold1_length_64517869:.... 218M of 1,073M

どれか1つの
染色体
scaffoldを選ぶ

ビューイングの拡大縮小

IGV

File Genomes View Tracks Regions Tools Help

buc.genome.fasta gn:buc gn:buc:225,621-234,301 Go

8,681 bp

226,000 bp 227,000 bp 228,000 bp 229,000 bp 230,000 bp 231,000 bp 232,000 bp 233,000 bp 234,000 bp

buc.gtf

BU205-RA BU206-RA BU207-RA BU208-RA BU209-RA BU210-RA BU211-RA BU212-RA BU213-RA

type: CDS
geneid: BU207
transcript_id: BU207-RA
gene_name: lpdA

カーソルのgene modelに
持っていくことで、gtfファイル
の記述がpop upされる。

トラックの移動
マウスのドラッグ移動が可能
← →による移動も可能
Feature trackを指定して、
Ctrl+F gene model単位で右に移動
Ctrl+B gene model単位で左に移動
Shift+Ctrl+F exon単位で右に移動

IGV

File Genomes View Tracks Regions Tools Help

buc.genome.fasta gn.buc gn.buc:225,621-234,301 Go

226,000 bp 227,000 bp 228,000 bp 229,000 bp 230,000 bp 231,000 bp 232,000 bp 233,000 bp 234,000 bp

8,681 bp

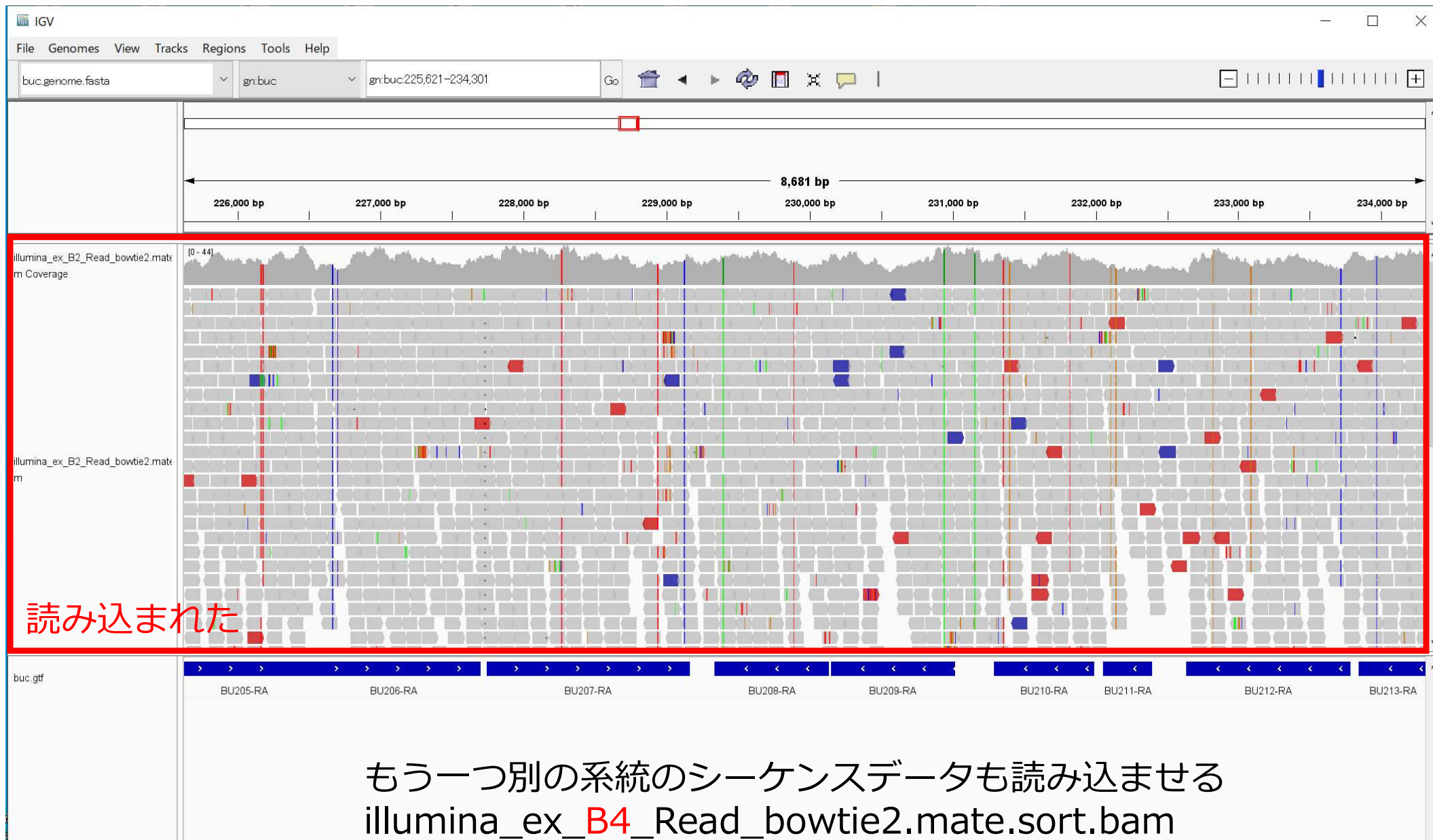
buc.gtf

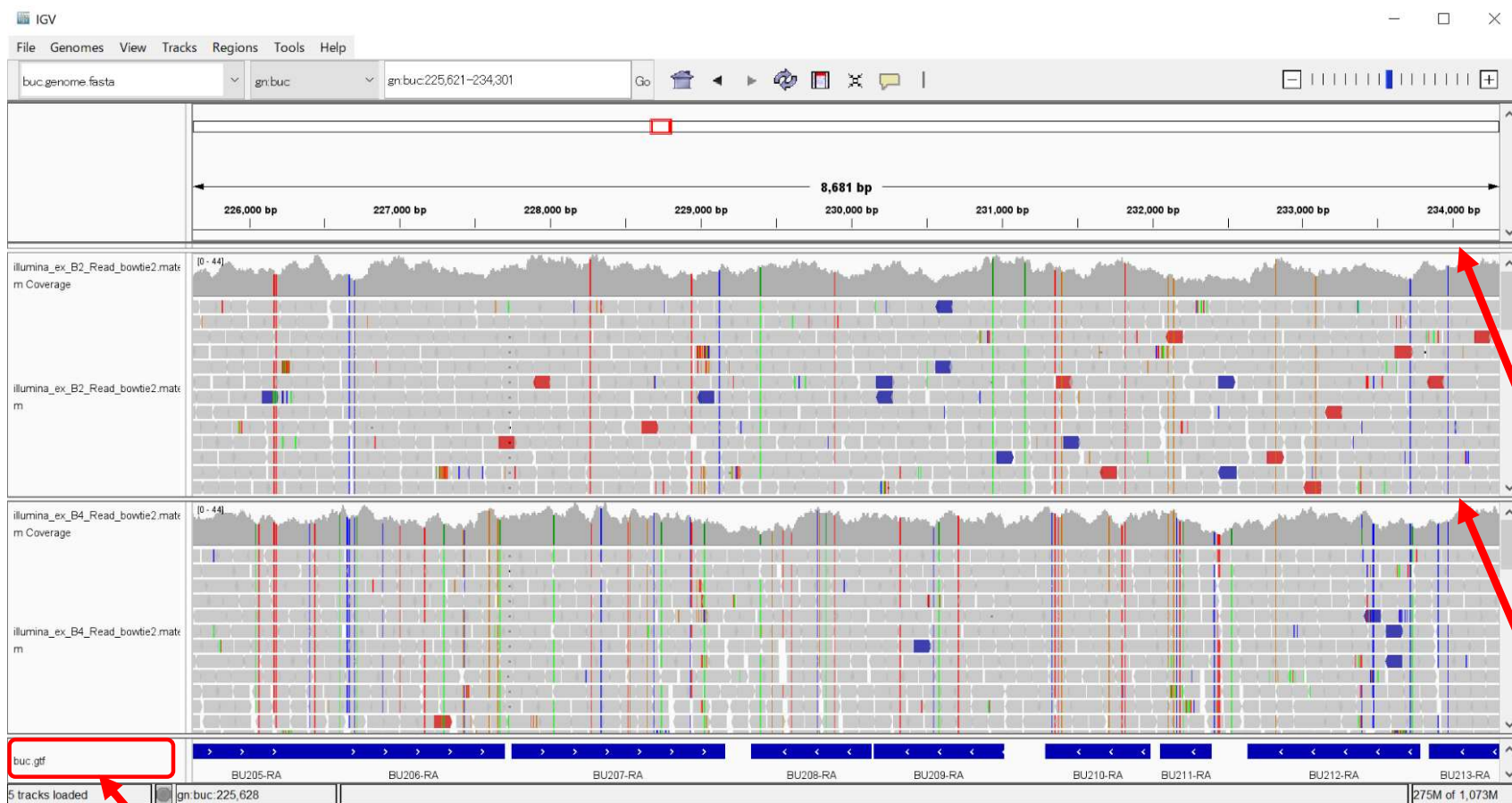
BU205-RA BU206-RA BU207-RA BU208-RA BU209-RA BU210-RA BU211-RA BU212-RA BU213-RA

シーケンスデータを読み込ませる
File -> Load from File

illumina_ex_B2_Read_bowtie2.mate.sort.bam

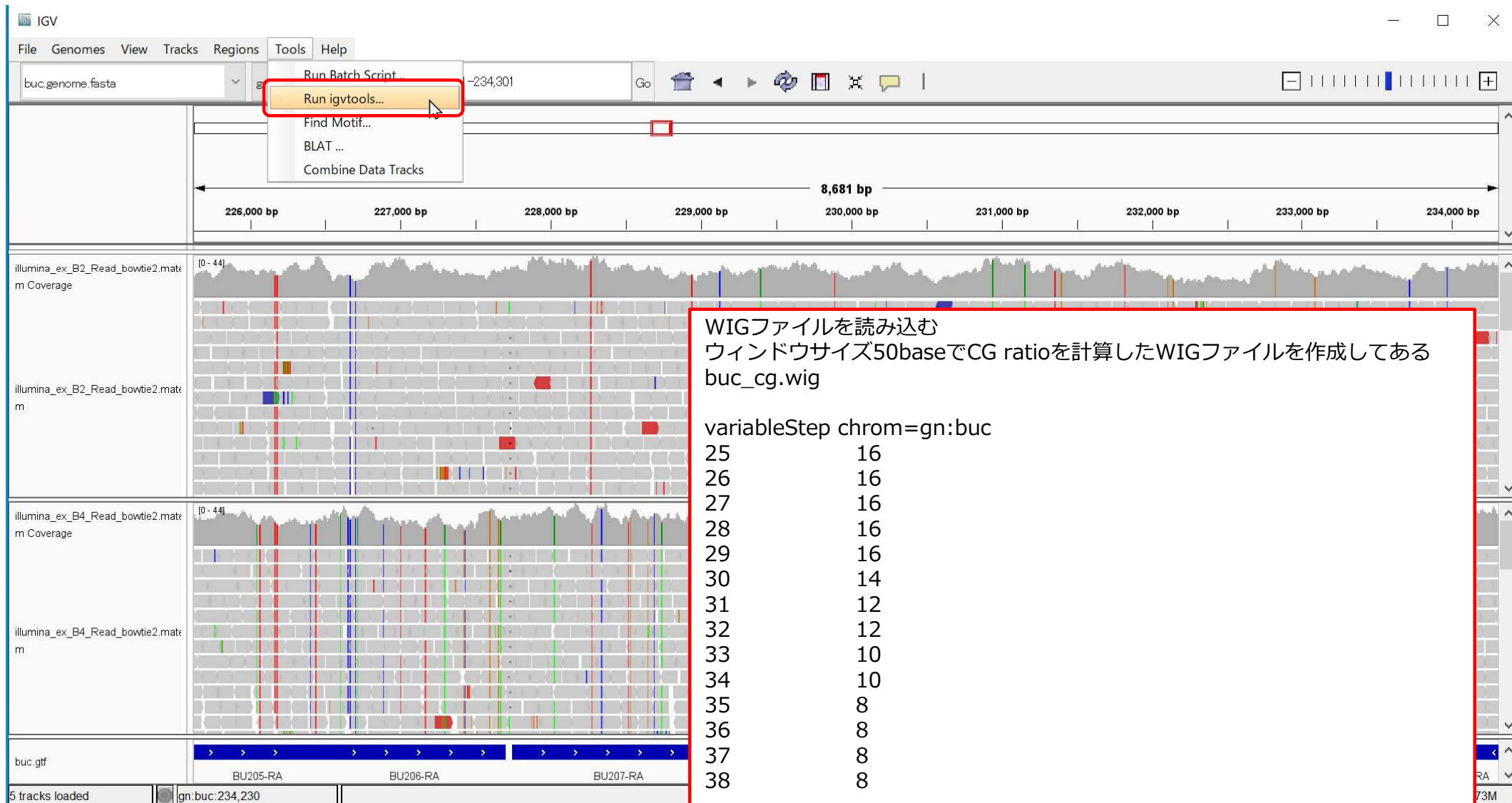
同ファイル名のindexファイル
illumina_ex_B2_Read_bowtie2.mate.sort.bam.bai
が同じフォルダーにないといけない。





各Trackはドラッグアンドドロップで移動できる

こういう所を
クリックしたまま
移動させることで
幅を変更



File -> Load from File でそのまま読み込ませることも可能だが、
テキストファイルで膨大なサイズだと非常に時間がかかる。
Tools -> Run igvtools

IGVtoolsを起動

Tools → Run igvtools

igvtools

Command toTDF

Input File D:\kyamaguc\Desktop\link\分析室\GITC_ゲノムインフォマティクストレーニングコース\トレーニングコース2020準備中\my_data_seq_2016spring\IGV\buc_cg wig Browse

Output File %kyamaguc\Desktop\link\分析室\GITC_ゲノムインフォマティクストレーニングコース\トレーニングコース2020準備中\my_data_seq_2016spring\IGV\buc_cg wig tdf Browse

Genome amaguc\Desktop\link\分析室\GITC_ゲノムインフォマティクストレーニングコース\トレーニングコース2020準備中\my_data_seq_2016spring\IGV\buc.genome.fasta Browse

TDF and Count options

Zoom Levels 7

Window Functions ☐ Min ☐ Max ☒ Mean ☐ Median ☐ 2% ☐ 10% ☐ 90% ☐ 98%

Probe to Loci Mapping Browse

Window Size 25

Extension Factor

☐ Count as Pairs

Sort Options

Temp Directory Browse

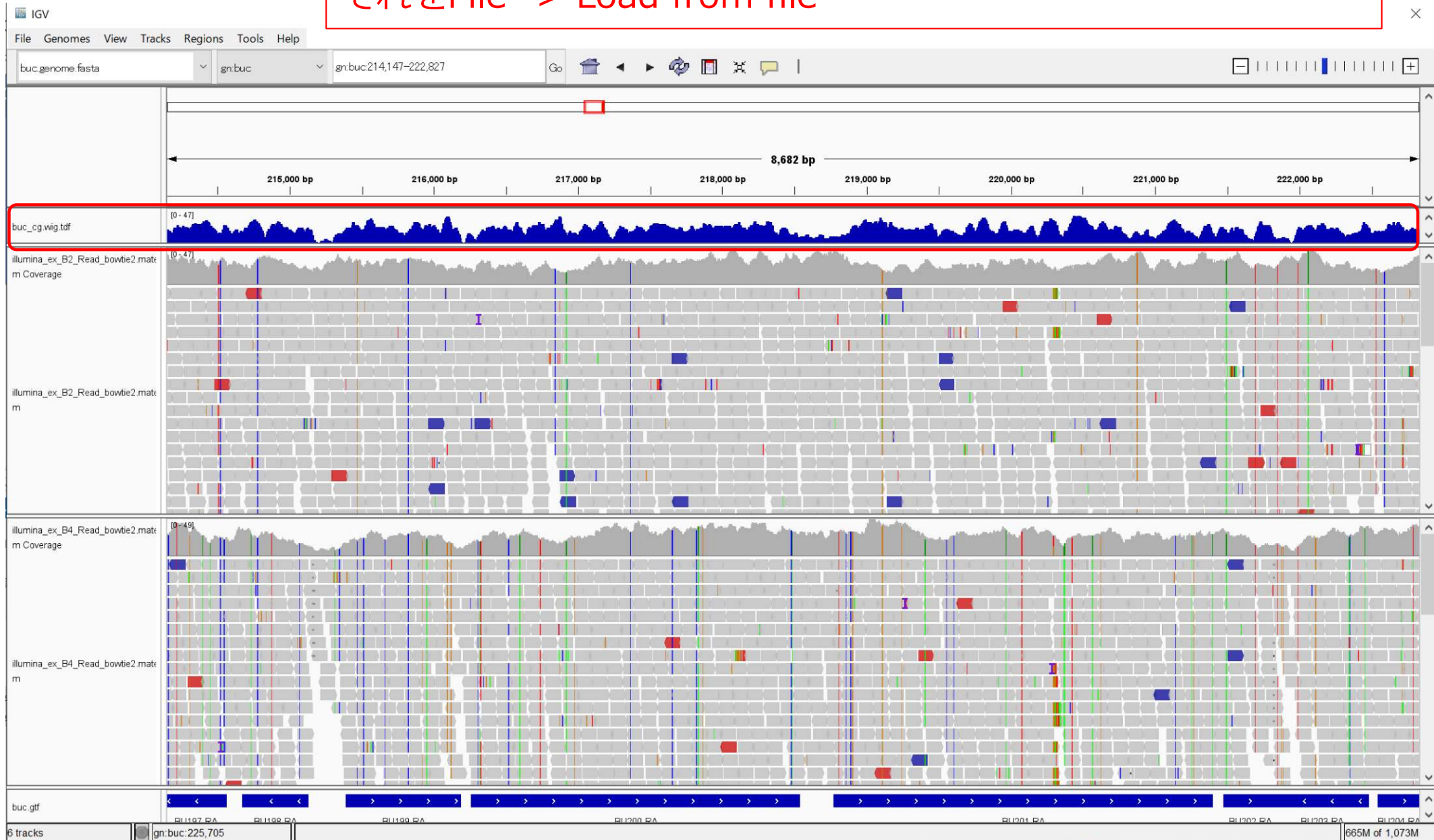
Max Records 500000

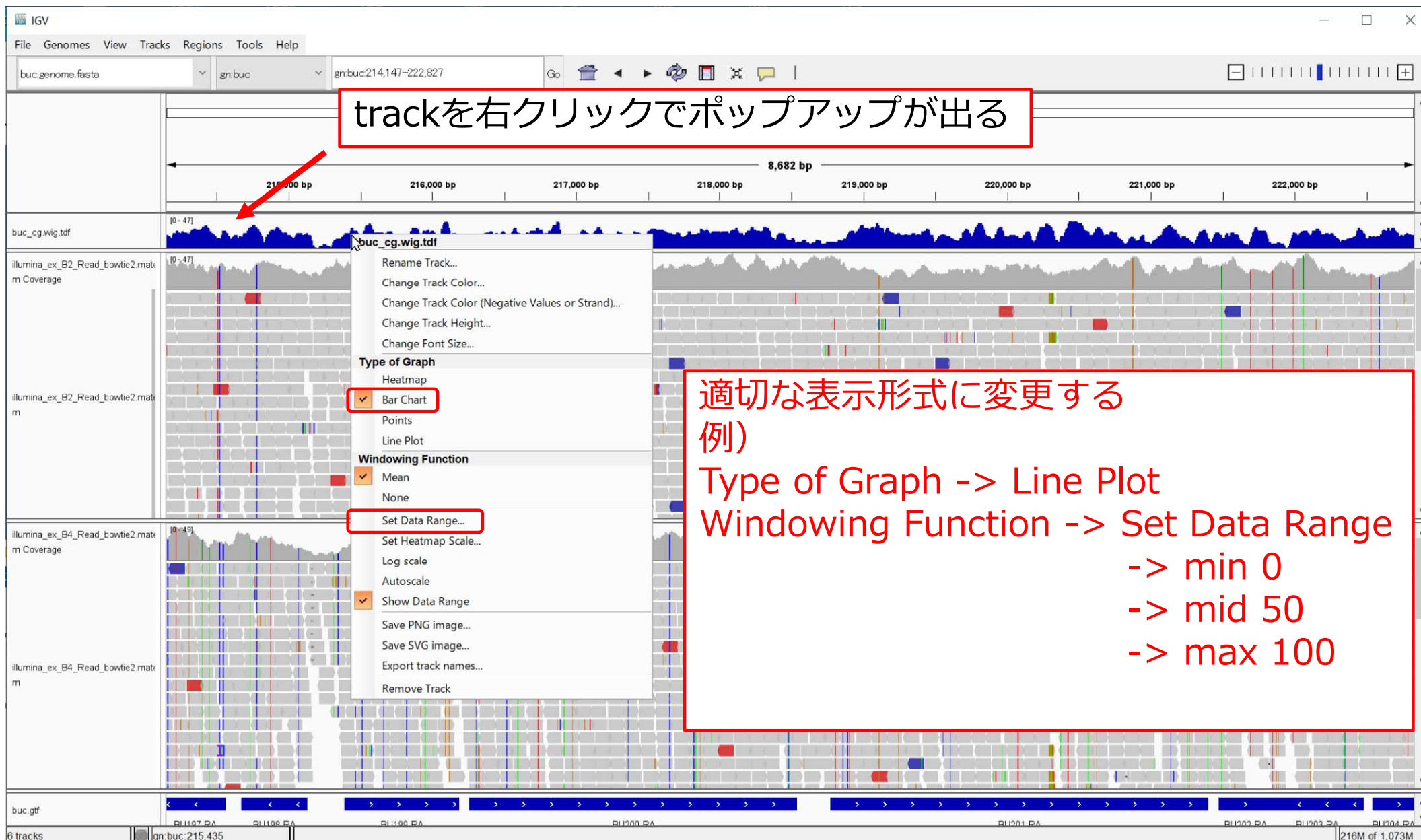
Close Run

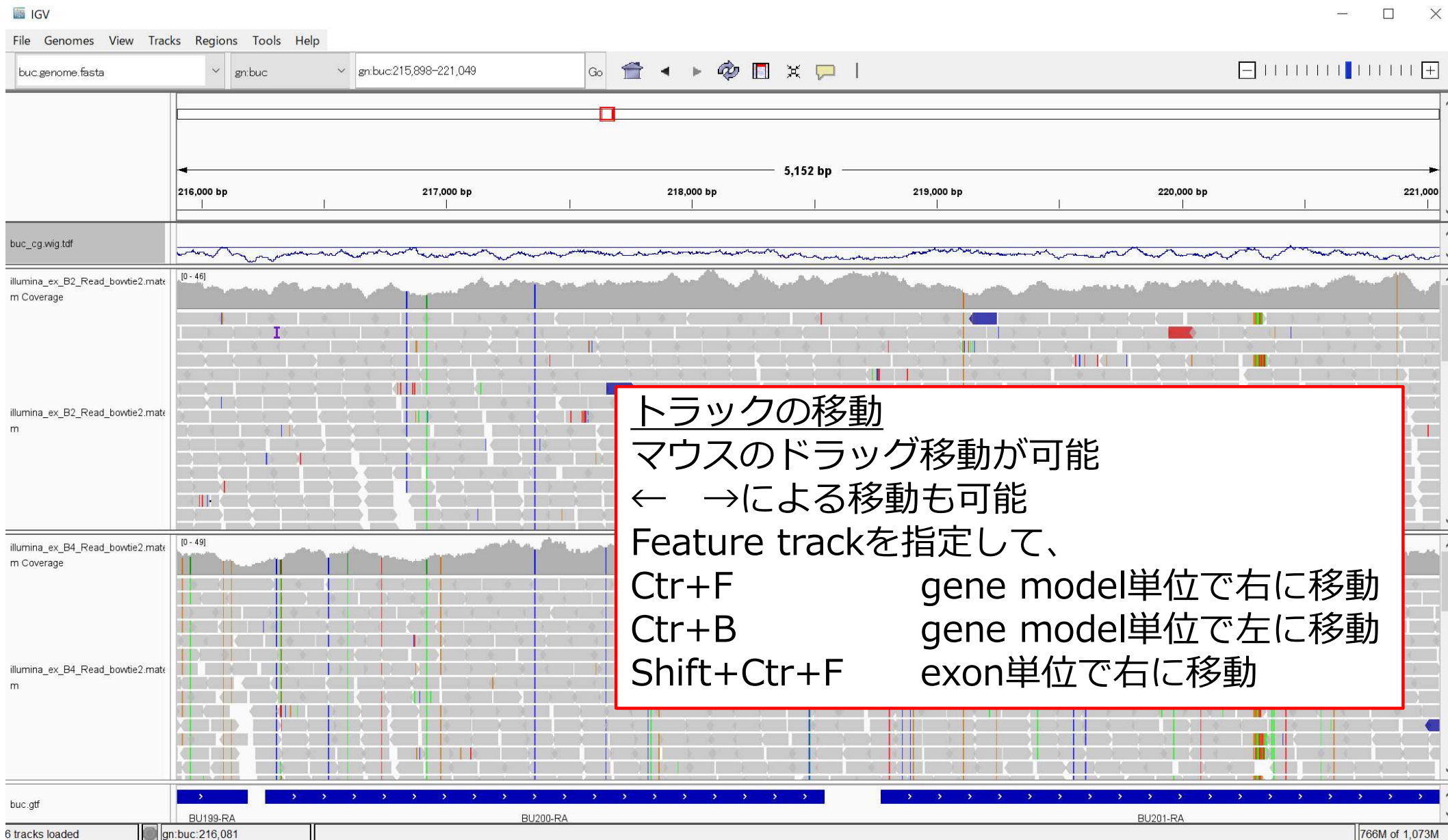
Messages

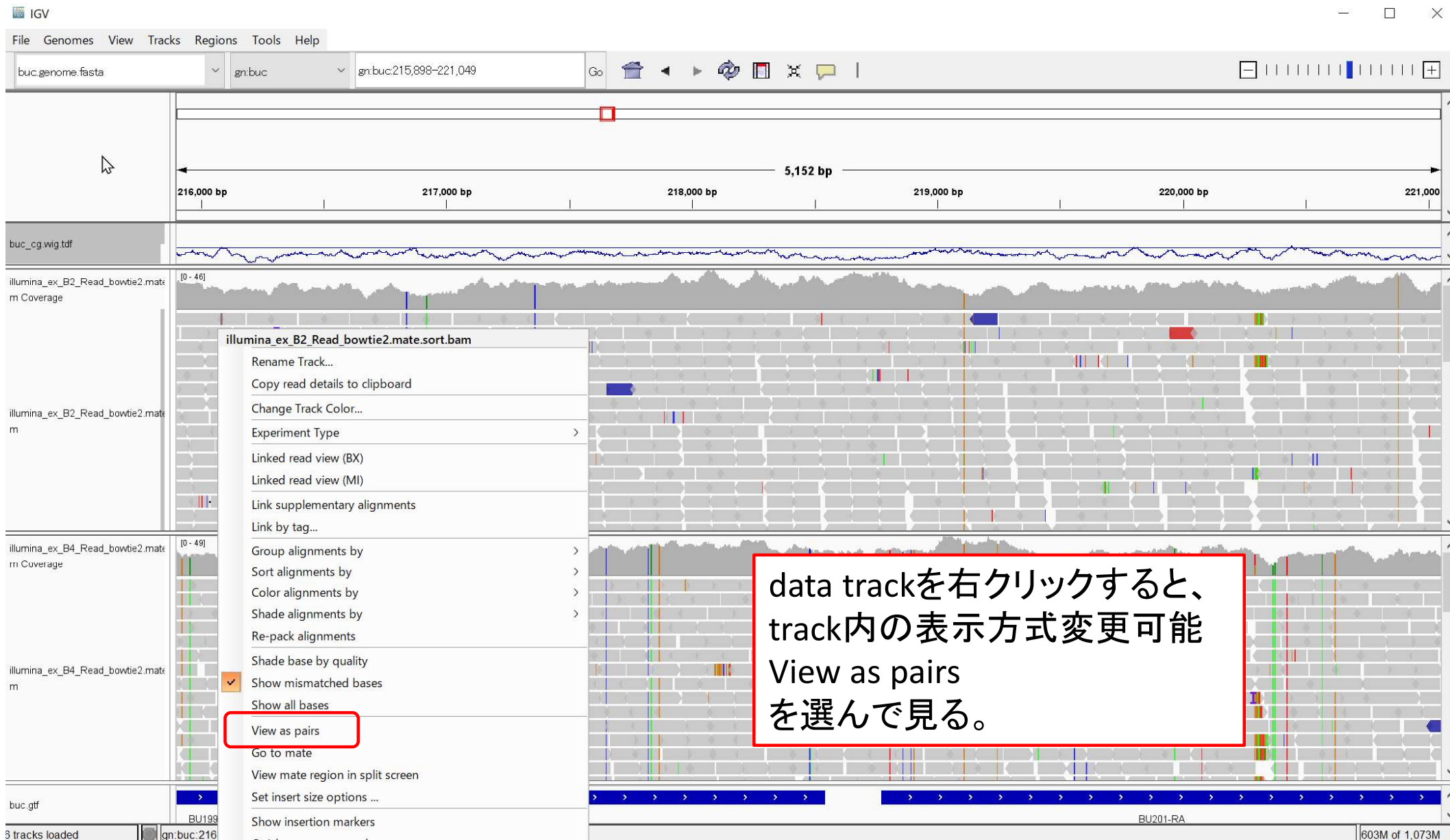
```
java.awt.event.ComponentEvent[COMPONENT_RESIZED (0,40 1124x737)] on org.broad.igv.ui.panel.MainPanel[0,40,1124x737,layout=java.awt.BorderLayout,alignmentX=0.0,a
java.awt.event.ComponentEvent[COMPONENT_RESIZED (0,40 1121x734)] on org.broad.igv.ui.panel.MainPanel[0,40,1121x734,layout=java.awt.BorderLayout,alignmentX=0.0,a
java.awt.event.ComponentEvent[COMPONENT_RESIZED (0,40 1119x731)] on org.broad.igv.ui.panel.MainPanel[0,40,1119x731,layout=java.awt.BorderLayout,alignmentX=0.0,a
java.awt.event.ComponentEvent[COMPONENT_RESIZED (0,40 1119x731)] on org.broad.igv.ui.panel.MainPanel[0,40,1119x731,layout=java.awt.BorderLayout,alignmentX=0.0,a
java.awt.event.ComponentEvent[COMPONENT_RESIZED (0,40 1115x727)] on org.broad.igv.ui.panel.MainPanel[0,40,1115x727,layout=java.awt.BorderLayout,alignmentX=0.0,a
java.awt.event.ComponentEvent[COMPONENT_RESIZED (0,40 1115x727)] on org.broad.igv.ui.panel.MainPanel[0,40,1115x727,layout=java.awt.BorderLayout,alignmentX=0.0,a
java.awt.event.ComponentEvent[COMPONENT_RESIZED (0,40 1108x722)] on org.broad.igv.ui.panel.MainPanel[0,40,1108x722,layout=java.awt.BorderLayout,alignmentX=0.0,a
java.awt.event.ComponentEvent[COMPONENT_RESIZED (0,40 1108x722)] on org.broad.igv.ui.panel.MainPanel[0,40,1108x722,layout=java.awt.BorderLayout,alignmentX=0.0,a
```

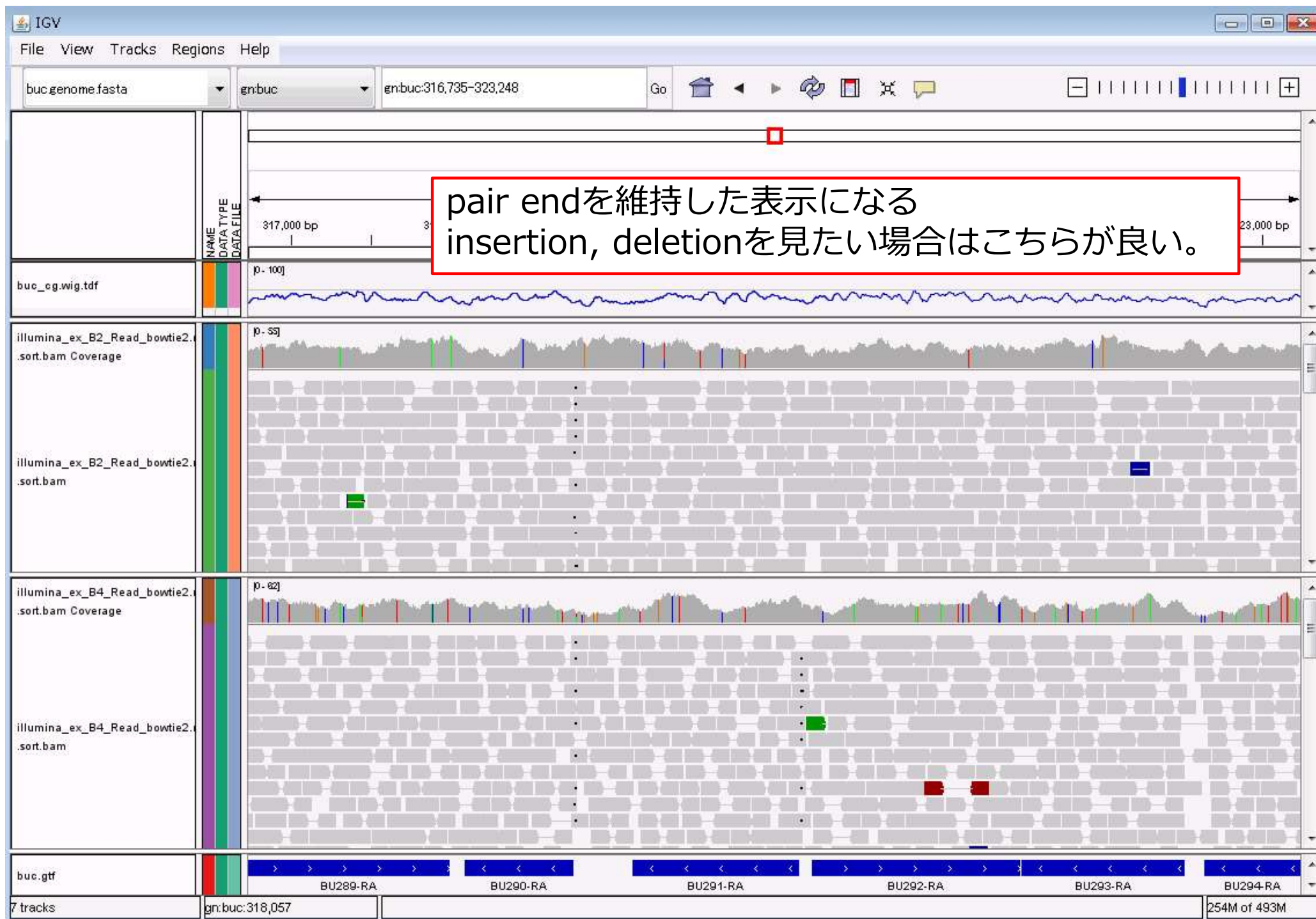
WIGファイルがバイナリー化されたtdfファイルができるので、
それをFile -> Load from file



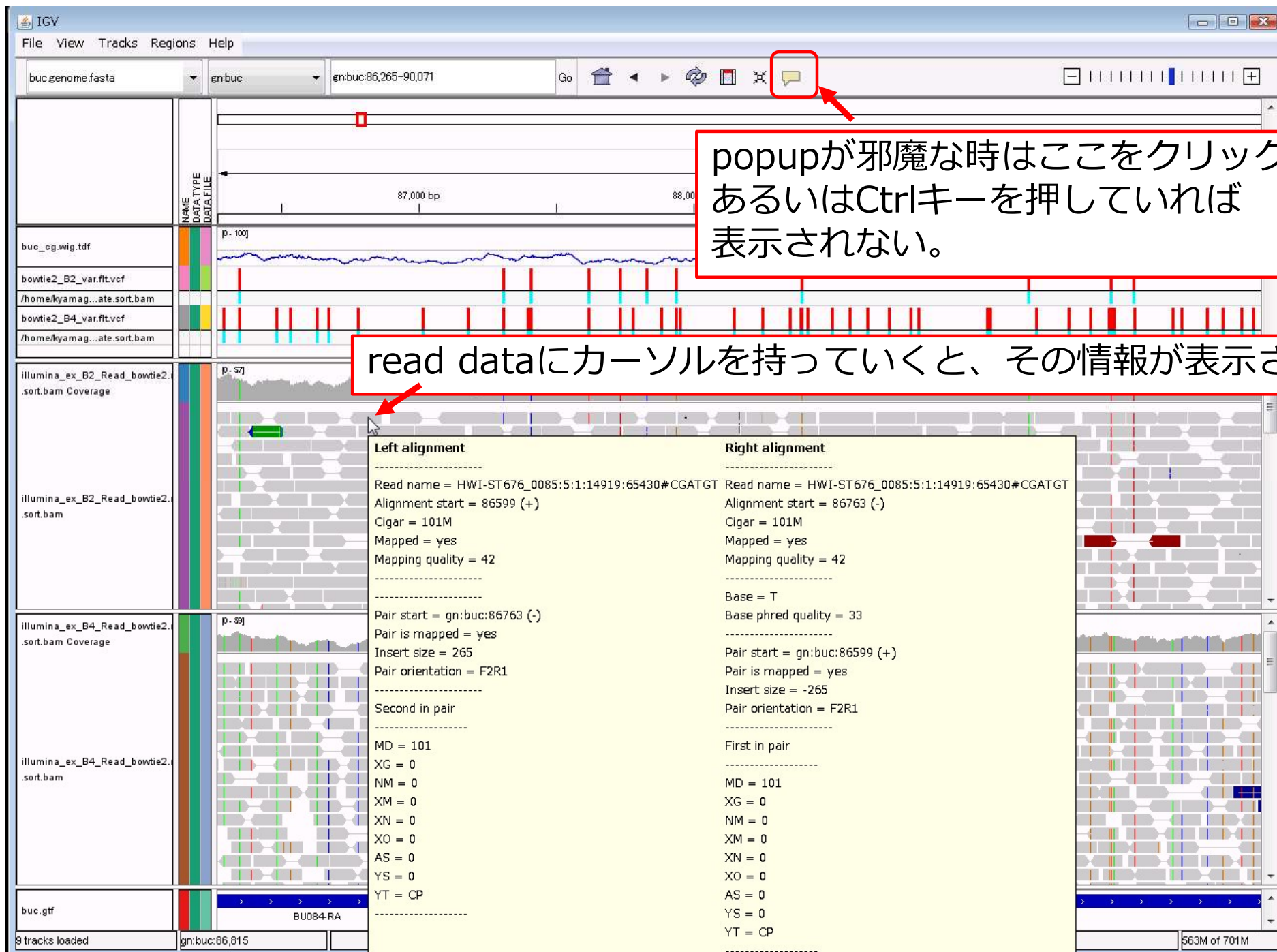












popupが邪魔な時はここをクリック、
あるいはCtrlキーを押していれば
表示されない。

read dataにカーソルを持ていくと、その情報が表示される

Gene prediction結果をIGVで見ながら考察する

2023.06.05

カブトムシのゲノムを解読し公開

自然科学研究機構 基礎生物学研究所
金沢大学

Linked read sequenceをsupernovaでアセンブル
Scaffold N50 8.02Mbのアセンブルを得た
実習用にscaffold27(15,574,170bp)のみ
取り出した

fastaファイル kabuto_part27.fasta

論文で報告している、short readによるRNA-Seq
と近縁種のタンパク質配列をベースにgene predictionした、
23,987 protein-coding sequenceを記述した

gffファイル p04_part27.gff

実習

これらをIGVを使って可視化してみよう

カブトムシアセンブル結果の以下fastaファイルを読み込む

File -> Load from File

kabuto_part27.fasta

protein-coding sequenceを記述したgffファイルを読み込む

File -> Load from File

p04_part27.gff

基礎生物学研究所の森田慎一助教、新美輝幸教授と重信秀治教授を中心として、金沢大学やモンタナ大学などが参加した国際共同研究チームは、カブトムシ *Trypoxylus dichotomus septentrionalis* の核ゲノムとミトコンドリアゲノム解読し、データベースを公開しました。

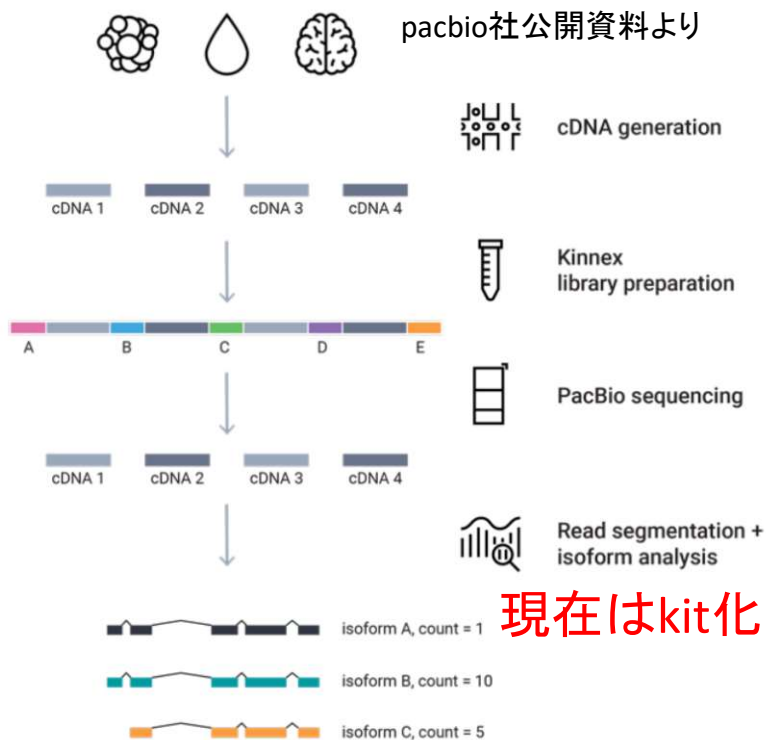
カブトムシの特徴である角がどのように獲得されたのかは、未だ明らかになっていません。詳細なゲノム情報は、角獲得の解明に対する鍵となると期待されています。今回研究チームは、次世代シーケンシング技術を用いてカブトムシのゲノムを解読し、角形成の遺伝的メカニズムを理解する基盤を確立しました。この新たなゲノム情報は、カブトムシの遺伝子機能や進化について包括的な解析を可能にし、これまでの研究を超える新たな可能性をもたらします。公開されたゲノム情報は、様々なカブトムシ研究の一助となることが期待されます。

本研究成果は科学学術誌 *Scientific Reports* に2023年5月30日付けで掲載されました。



全長cDNA配列をハイスループットで得る新技術

全長cDNAをそのままシーケンスするISO-Seq法
さらにcDNAを連結して、ハイスループットにシーケンス



現在はkit化された製品が利用可能

Figure 2. Kinnex full-length RNA sequencing. Full-length cDNA molecules are concatenated into a large-insert library, sequenced, then processed using the PacBio software.

short readからのアセンブルで生じるミスアセンブルを排除
オルタナティブスプライシング状況が明確に分かる
隣接する遺伝子を明確に区分け

nature biotechnology

Brief Communication

<https://doi.org/10.1038/s41587-023-01815-7>

High-throughput RNA isoform sequencing using programmed cDNA concatenation

Received: 1 October 2021

Accepted: 2 May 2023

Published online: 08 June 2023

Check for updates

Aziz M. Al'Khafaji^{1,11}✉, Jonathan T. Smith^{1,11}, Kiran V. Garimella^{1,11}✉, Mehrtash Babadi^{1,11}✉, Victoria Popic^{1,11}✉, Moshe Sade-Feldman^{1,2}, Michael Gatzen¹, Siranush Sarkizova¹, Marc A. Schwartz^{1,3,4,5}, Emily M. Blaum^{1,2}, Allyson Day¹, Maura Costello¹, Tera Bowers¹, Stacey Gabriel¹, Eric Banks¹, Anthony A. Philippakis¹, Genevieve M. Boland⁶, Paul C. Blainey^{1,7,8}✉ & Nir Hacohen^{1,2,9,10}✉

Full-length RNA-sequencing methods using long-read technologies can capture complete transcript isoforms, but their throughput is limited. We introduce multiplexed arrays isoform sequencing (MAS-ISO-seq), a technique for programmably concatenating complementary DNAs (cDNAs) into molecules optimal for long-read sequencing, increasing the throughput >15-fold to nearly 40 million cDNA reads per run on the Sequel IIe sequencer. When applied to single-cell RNA sequencing of tumor-infiltrating T cells, MAS-ISO-seq demonstrated a 12- to 32-fold increase in the discovery of differentially spliced genes.

この結果をゲノムにmappingした
bamファイル mapped_part27.bam

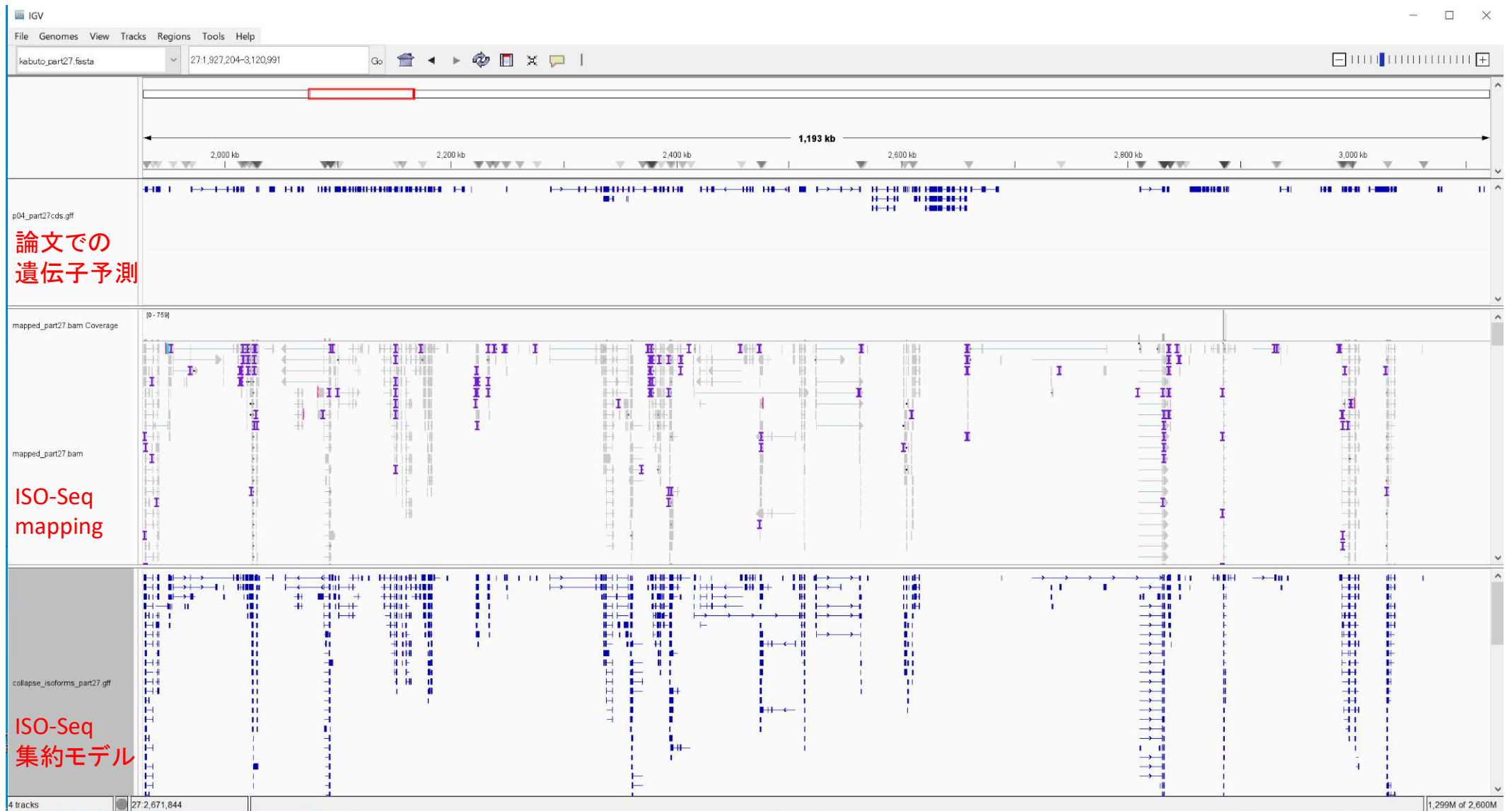
同一配列を集約した
gffファイル collapse_isoforms_part27.gff

を用意した

実習
これら2つのファイルを追加しよう

File -> Load from File
mapped_part27.bam

File -> Load from File
collapse_isoforms_part27.gff



可視化によって直感的な比較検討が可能に。

IGV紹介のまとめ



可視化ツールとして十分な機能を持つ

- ・ 無料
- ・ 比較的簡単・お手軽
- ・ 自分で見るためにもよし、人に見せるためにもよし
- ・ 利用範囲は次世代DNAシーケンサーに限定しない
広くゲノミクスの解析に有用

ごく一部のみの機能を紹介しました。
ウェブサイトを見ながら復習をお勧めします。